

EXERCISE 3A.2

1 Simplify:

a $\sqrt{2} + \sqrt{2}$

b $\sqrt{2} - \sqrt{2}$

c $3\sqrt{2} - 2\sqrt{2}$

d $2\sqrt{3} - \sqrt{3}$

e $5\sqrt{7} + 2\sqrt{7}$

f $3\sqrt{5} - 6\sqrt{5}$

g $3\sqrt{2} + 4\sqrt{2} - \sqrt{2}$

h $6\sqrt{2} - 9\sqrt{2}$

i $\sqrt{5} + 7\sqrt{5}$

j $3\sqrt{2} - 5\sqrt{2} - \sqrt{2}$

k $3\sqrt{3} - \sqrt{3} + 2\sqrt{3}$

l $3\sqrt{5} + 7\sqrt{5} - 10$

2 Simplify:

a $3\sqrt{2} + 2\sqrt{3} - \sqrt{2} + 5\sqrt{3}$

b $7\sqrt{2} - 4\sqrt{3} - 2\sqrt{2} + 3\sqrt{3}$

c $-6\sqrt{2} - 2\sqrt{3} - \sqrt{2} + 6\sqrt{3}$

d $2\sqrt{5} + 4\sqrt{2} + 9\sqrt{5} - 9\sqrt{2}$

e $3\sqrt{2} - 5\sqrt{7} - \sqrt{2} - 5\sqrt{7}$

f $3\sqrt{2} + 4\sqrt{11} + 6 - \sqrt{2} - \sqrt{11} - 3$

g $6\sqrt{6} - 2\sqrt{2} - \sqrt{2} - 5\sqrt{6} + 4$

h $5\sqrt{3} - 6\sqrt{7} - 5 + 4\sqrt{3} + \sqrt{7} - 8$

EXERCISE 3B.1

1 Simplify:

a $\sqrt{2} \times \sqrt{5}$

b $\sqrt{3} \times \sqrt{7}$

c $\sqrt{3} \times \sqrt{11}$

d $\sqrt{7} \times \sqrt{7}$

e $\sqrt{3} \times 2\sqrt{3}$

f $2\sqrt{2} \times \sqrt{5}$

g $3\sqrt{3} \times 2\sqrt{2}$

h $2\sqrt{3} \times 3\sqrt{5}$

i $\sqrt{2} \times \sqrt{3} \times \sqrt{5}$

j $\sqrt{3} \times \sqrt{2} \times 2\sqrt{2}$

k $-3\sqrt{2} \times (\sqrt{2})^3$

l $(3\sqrt{2})^3 \times (\sqrt{3})^3$

2 Simplify:

a $\frac{\sqrt{8}}{\sqrt{2}}$

b $\frac{\sqrt{2}}{\sqrt{8}}$

c $\frac{\sqrt{18}}{\sqrt{2}}$

d $\frac{\sqrt{2}}{\sqrt{18}}$

e $\frac{\sqrt{20}}{\sqrt{5}}$

f $\frac{\sqrt{5}}{\sqrt{20}}$

g $\frac{\sqrt{27}}{\sqrt{3}}$

h $\frac{\sqrt{18}}{\sqrt{3}}$

i $\frac{\sqrt{3}}{\sqrt{30}}$

j $\frac{\sqrt{50}}{\sqrt{2}}$

k $\frac{2\sqrt{6}}{\sqrt{24}}$

l $\frac{5\sqrt{75}}{\sqrt{3}}$

3 Write the following in the form $k\sqrt{2}$:

a $\sqrt{8}$

b $\sqrt{18}$

c $\sqrt{50}$

d $\sqrt{98}$

e $\sqrt{200}$

f $\sqrt{288}$

g $\sqrt{20\,000}$

h $\sqrt{\frac{1}{2}}$

4 Write the following in the form $k\sqrt{3}$:

a $\sqrt{12}$

b $\sqrt{27}$

c $\sqrt{75}$

d $\sqrt{\frac{1}{3}}$

5 Write the following in the form $k\sqrt{5}$:

a $\sqrt{20}$

b $\sqrt{45}$

c $\sqrt{125}$

d $\sqrt{\frac{1}{5}}$

6 **a** Find:

i $\sqrt{16} + \sqrt{9}$

ii $\sqrt{16 + 9}$

iii $\sqrt{25} - \sqrt{9}$

iv $\sqrt{25 - 9}$

b i Does $\sqrt{16 + 9} = \sqrt{16} + \sqrt{9}$?

ii Does $\sqrt{25 - 9} = \sqrt{25} - \sqrt{9}$?

c Copy and complete: In general $\sqrt{a + b} \neq \dots\dots$ and $\sqrt{a - b} \neq \dots\dots$

EXERCISE 3B.2

1 Write the following in simplest radical form:

a $\sqrt{24}$

b $\sqrt{50}$

c $\sqrt{54}$

d $\sqrt{40}$

e $\sqrt{56}$

f $\sqrt{63}$

g $\sqrt{52}$

h $\sqrt{44}$

i $\sqrt{60}$

j $\sqrt{90}$

k $\sqrt{96}$

l $\sqrt{68}$

m $\sqrt{175}$

n $\sqrt{162}$

o $\sqrt{128}$

p $\sqrt{700}$

2 Write the following in simplest radical form:

a $\sqrt{\frac{5}{9}}$

b $\sqrt{\frac{18}{4}}$

c $\sqrt{\frac{12}{16}}$

d $\sqrt{\frac{75}{36}}$

EXERCISE 3D

1 Express with integer denominator:

a $\frac{1}{\sqrt{2}}$

b $\frac{2}{\sqrt{2}}$

c $\frac{4}{\sqrt{2}}$

d $\frac{10}{\sqrt{2}}$

e $\frac{\sqrt{7}}{\sqrt{2}}$

f $\frac{1}{\sqrt{3}}$

g $\frac{3}{\sqrt{3}}$

h $\frac{4}{\sqrt{3}}$

i $\frac{18}{\sqrt{3}}$

j $\frac{\sqrt{11}}{\sqrt{3}}$

k $\frac{1}{\sqrt{5}}$

l $\frac{3}{\sqrt{5}}$

m $\frac{\sqrt{3}}{\sqrt{5}}$

n $\frac{15}{\sqrt{5}}$

o $\frac{125}{\sqrt{5}}$

p $\frac{\sqrt{10}}{\sqrt{2}}$

q $\frac{1}{2\sqrt{3}}$

r $\frac{2\sqrt{2}}{\sqrt{3}}$

s $\frac{15}{2\sqrt{5}}$

t $\frac{1}{(\sqrt{2})^3}$

2 Express with integer denominator:

a $\frac{1}{3 - \sqrt{5}}$

b $\frac{1}{2 + \sqrt{3}}$

c $\frac{1}{4 - \sqrt{11}}$

d $\frac{\sqrt{2}}{5 + \sqrt{2}}$

e $\frac{\sqrt{3}}{3 + \sqrt{3}}$

f $\frac{5}{2 - 3\sqrt{2}}$

g $\frac{-\sqrt{5}}{3 + 2\sqrt{5}}$

h $\frac{3 - \sqrt{7}}{2 + \sqrt{7}}$

3 Write in the form $a + b\sqrt{2}$ where $a, b \in \mathbb{Q}$:

a $\frac{4}{2 - \sqrt{2}}$

b $\frac{-5}{1 + \sqrt{2}}$

c $\frac{1 - \sqrt{2}}{1 + \sqrt{2}}$

d $\frac{\sqrt{2} - 2}{3 - \sqrt{2}}$

e $\frac{\frac{1}{\sqrt{2}}}{1 - \frac{1}{\sqrt{2}}}$

f $\frac{1 + \frac{1}{\sqrt{2}}}{1 - \frac{1}{\sqrt{2}}}$

g $\frac{1}{1 - \frac{\sqrt{2}}{3}}$

h $\frac{\frac{\sqrt{2}}{2} + 1}{1 - \frac{\sqrt{2}}{4}}$

EXERCISE 3C

1 Expand and simplify:

a $\sqrt{2}(\sqrt{5} + \sqrt{2})$

b $\sqrt{2}(3 - \sqrt{2})$

c $\sqrt{3}(\sqrt{3} + 1)$

d $\sqrt{3}(1 - \sqrt{3})$

e $\sqrt{7}(7 - \sqrt{7})$

f $\sqrt{5}(2 - \sqrt{5})$

g $\sqrt{11}(2\sqrt{11} - 1)$

h $\sqrt{6}(1 - 2\sqrt{6})$

i $\sqrt{3}(\sqrt{3} + \sqrt{2} - 1)$

j $2\sqrt{3}(\sqrt{3} - \sqrt{5})$

k $2\sqrt{5}(3 - \sqrt{5})$

l $3\sqrt{5}(2\sqrt{5} + \sqrt{2})$

2 Expand and simplify:

a $-\sqrt{2}(3 - \sqrt{2})$

b $-\sqrt{2}(\sqrt{2} + \sqrt{3})$

c $-\sqrt{2}(4 - \sqrt{2})$

d $-\sqrt{3}(1 + \sqrt{3})$

e $-\sqrt{3}(\sqrt{3} + 2)$

f $-\sqrt{5}(2 + \sqrt{5})$

g $-(\sqrt{2} + 3)$

h $-\sqrt{5}(\sqrt{5} - 4)$

i $-(3 - \sqrt{7})$

j $-\sqrt{11}(2 - \sqrt{11})$

k $-(\sqrt{3} - \sqrt{7})$

l $-2\sqrt{2}(1 - \sqrt{2})$

m $-3\sqrt{3}(5 - \sqrt{3})$

n $-7\sqrt{2}(\sqrt{2} + \sqrt{6})$

o $(-\sqrt{2})^3(3 - \sqrt{2})$

3 Expand and simplify:

a $(1 + \sqrt{2})(2 + \sqrt{2})$

b $(2 + \sqrt{3})(2 + \sqrt{3})$

c $(\sqrt{3} + 2)(\sqrt{3} - 1)$

d $(4 - \sqrt{2})(3 + \sqrt{2})$

e $(1 + \sqrt{3})(1 - \sqrt{3})$

f $(5 + \sqrt{7})(2 - \sqrt{7})$

g $(\sqrt{5} + 2)(\sqrt{5} - 3)$

h $(\sqrt{7} - \sqrt{3})(\sqrt{7} + \sqrt{3})$

i $(2\sqrt{2} + \sqrt{3})(2\sqrt{2} - \sqrt{3})$

j $(4 - \sqrt{2})(3 - \sqrt{2})$

4 Expand and simplify:

a $(1 + \sqrt{2})^2$

b $(2 - \sqrt{3})^2$

c $(\sqrt{3} + 2)^2$

d $(1 + \sqrt{5})^2$

e $(\sqrt{2} - \sqrt{3})^2$

f $(5 - \sqrt{2})^2$

g $(\sqrt{2} + \sqrt{7})^2$

h $(4 - \sqrt{6})^2$

i $(\sqrt{6} - \sqrt{2})^2$

j $(\sqrt{5} + 2\sqrt{2})^2$

k $(\sqrt{5} - 2\sqrt{2})^2$

l $(6 + \sqrt{8})^2$

m $(5\sqrt{2} - 1)^2$

n $(3 - 2\sqrt{2})^2$

o $(1 + 3\sqrt{2})^2$

5 Expand and simplify:

a $(4 + \sqrt{3})(4 - \sqrt{3})$

b $(5 - \sqrt{2})(5 + \sqrt{2})$

c $(\sqrt{5} - 2)(\sqrt{5} + 2)$

d $(\sqrt{7} + 4)(\sqrt{7} - 4)$

e $(3\sqrt{2} + 2)(3\sqrt{2} - 2)$

f $(2\sqrt{5} - 1)(2\sqrt{5} + 1)$

g $(5 - 3\sqrt{3})(5 + 3\sqrt{3})$

h $(2 - 4\sqrt{2})(2 + 4\sqrt{2})$

i $(1 + 5\sqrt{7})(1 - 5\sqrt{7})$

6 Expand and simplify:

a $(\sqrt{3} + \sqrt{2})(\sqrt{3} - \sqrt{2})$ **b** $(\sqrt{7} + \sqrt{11})(\sqrt{7} - \sqrt{11})$ **c** $(\sqrt{x} - \sqrt{y})(\sqrt{y} + \sqrt{x})$

Harder Questions:

4 Write in simplest form:

a $\frac{1 + \sqrt{2}}{1 - \sqrt{2}} + \frac{1 - \sqrt{2}}{1 + \sqrt{2}}$

b $\frac{2 + \sqrt{5}}{2 - \sqrt{5}} - \frac{\sqrt{5}}{2 + \sqrt{5}}$

c $\frac{4 - \sqrt{3}}{3 - 2\sqrt{2}} - \frac{2\sqrt{3}}{3 + 2\sqrt{2}}$

5 Find $\sqrt{\frac{3 + 2\sqrt{2}}{3 - 2\sqrt{2}}}$ giving your answer in the form $a + b\sqrt{2}$ where $a, b \in \mathbb{Q}$.

6 If $\frac{1}{\sqrt{3}} - \frac{1}{\sqrt{2}} = p$, find $\sqrt{6}$ in terms of p .

7 If $x = \sqrt{5} - \sqrt{3}$, find x^2 and hence show that $x^4 - 16x^2 + 4 = 0$.

You have just shown that one solution of $x^4 - 16x^2 + 4 = 0$ is $x = \sqrt{5} - \sqrt{3}$.

8 If $u_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1 + \sqrt{5}}{2} \right)^n - \left(\frac{1 - \sqrt{5}}{2} \right)^n \right]$, evaluate u_n for $n = 1, 2, 3$ and 4 .

Solutions:

EXERCISE 3A.2

1 **a** $2\sqrt{2}$ **b** 0 **c** $\sqrt{2}$ **d** $\sqrt{3}$ **e** $7\sqrt{7}$ **f** $-3\sqrt{5}$
g $6\sqrt{2}$ **h** $-3\sqrt{2}$ **i** $8\sqrt{5}$ **j** $-3\sqrt{2}$ **k** $4\sqrt{3}$
l $10\sqrt{5} - 10$

2 **a** $2\sqrt{2} + 7\sqrt{3}$ **b** $5\sqrt{2} - \sqrt{3}$ **c** $-7\sqrt{2} + 4\sqrt{3}$
d $11\sqrt{5} - 5\sqrt{2}$ **e** $2\sqrt{2} - 10\sqrt{7}$ **f** $2\sqrt{2} + 3\sqrt{11} + 3$
g $\sqrt{6} - 3\sqrt{2} + 4$ **h** $9\sqrt{3} - 5\sqrt{7} - 13$

Solutions:

EXERCISE 3B.1

- 1** **a** $\sqrt{10}$ **b** $\sqrt{21}$ **c** $\sqrt{33}$ **d** 7
 e 6 **f** $2\sqrt{10}$ **g** $6\sqrt{6}$ **h** $6\sqrt{15}$
 i $\sqrt{30}$ **j** $4\sqrt{3}$ **k** -12 **l** $162\sqrt{6}$
- 2** **a** 2 **b** $\frac{1}{2}$ **c** 3 **d** $\frac{1}{3}$ **e** 2 **f** $\frac{1}{2}$
 g 3 **h** $\sqrt{6}$ **i** $\frac{1}{\sqrt{10}}$ **j** 5 **k** 1 **l** 25
- 3** **a** $2\sqrt{2}$ **b** $3\sqrt{2}$ **c** $5\sqrt{2}$ **d** $7\sqrt{2}$
 e $10\sqrt{2}$ **f** $12\sqrt{2}$ **g** $100\sqrt{2}$ **h** $\frac{1}{2}\sqrt{2}$
- 4** **a** $2\sqrt{3}$ **b** $3\sqrt{3}$ **c** $5\sqrt{3}$ **d** $\frac{1}{3}\sqrt{3}$
- 5** **a** $2\sqrt{5}$ **b** $3\sqrt{5}$ **c** $5\sqrt{5}$ **d** $\frac{1}{5}\sqrt{5}$
- 6** **a** **i** 7 **ii** 5 **iii** 2 **iv** 4 **b** **i** no **ii** no
 c $\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$ and $\sqrt{a-b} \neq \sqrt{a} - \sqrt{b}$

EXERCISE 3B.2

- 1** **a** $2\sqrt{6}$ **b** $5\sqrt{2}$ **c** $3\sqrt{6}$ **d** $2\sqrt{10}$ **e** $2\sqrt{14}$ **f** $3\sqrt{7}$
 g $2\sqrt{13}$ **h** $2\sqrt{11}$ **i** $2\sqrt{15}$ **j** $3\sqrt{10}$ **k** $4\sqrt{6}$ **l** $2\sqrt{17}$
 m $5\sqrt{7}$ **n** $9\sqrt{2}$ **o** $8\sqrt{2}$ **p** $10\sqrt{7}$
- 2** **a** $\frac{\sqrt{5}}{3}$ **b** $\frac{3\sqrt{2}}{2}$ **c** $\frac{\sqrt{3}}{2}$ **d** $\frac{5\sqrt{3}}{6}$

EXERCISE 3D

1 **a** $\frac{\sqrt{2}}{2}$ **b** $\sqrt{2}$ **c** $2\sqrt{2}$ **d** $5\sqrt{2}$ **e** $\frac{\sqrt{14}}{2}$
f $\frac{\sqrt{3}}{3}$ **g** $\sqrt{3}$ **h** $\frac{4\sqrt{3}}{3}$ **i** $6\sqrt{3}$ **j** $\frac{\sqrt{33}}{3}$
k $\frac{\sqrt{5}}{5}$ **l** $\frac{3\sqrt{5}}{5}$ **m** $\frac{\sqrt{15}}{5}$ **n** $3\sqrt{5}$ **o** $25\sqrt{5}$
p $\sqrt{5}$ **q** $\frac{\sqrt{3}}{6}$ **r** $\frac{2\sqrt{6}}{3}$ **s** $\frac{3\sqrt{5}}{2}$ **t** $\frac{\sqrt{2}}{4}$

2 **a** $\frac{3 + \sqrt{5}}{4}$ **b** $2 - \sqrt{3}$ **c** $\frac{4 + \sqrt{11}}{5}$ **d** $\frac{5\sqrt{2} - 2}{23}$
e $\frac{\sqrt{3} - 1}{2}$ **f** $\frac{10 + 15\sqrt{2}}{-14}$ **g** $\frac{3\sqrt{5} - 10}{11}$ **h** $\frac{5\sqrt{7} - 13}{3}$

3 **a** $4 + 2\sqrt{2}$ **b** $5 - 5\sqrt{2}$ **c** $-3 + 2\sqrt{2}$ **d** $-\frac{4}{7} + \frac{1}{7}\sqrt{2}$
e $1 + \sqrt{2}$ **f** $3 + 2\sqrt{2}$ **g** $\frac{9}{7} + \frac{3}{7}\sqrt{2}$ **h** $\frac{10}{7} + \frac{6}{7}\sqrt{2}$

EXERCISE 3C

- 1** **a** $\sqrt{10} + 2$ **b** $3\sqrt{2} - 2$ **c** $3 + \sqrt{3}$ **d** $\sqrt{3} - 3$
e $7\sqrt{7} - 7$ **f** $2\sqrt{5} - 5$ **g** $22 - \sqrt{11}$ **h** $\sqrt{6} - 12$
i $3 + \sqrt{6} - \sqrt{3}$ **j** $6 - 2\sqrt{15}$ **k** $6\sqrt{5} - 10$ **l** $30 + 3\sqrt{10}$
- 2** **a** $2 - 3\sqrt{2}$ **b** $-2 - \sqrt{6}$ **c** $2 - 4\sqrt{2}$ **d** $-3 - \sqrt{3}$
e $-3 - 2\sqrt{3}$ **f** $-5 - 2\sqrt{5}$ **g** $-3 - \sqrt{2}$ **h** $-5 + 4\sqrt{5}$
i $\sqrt{7} - 3$ **j** $11 - 2\sqrt{11}$ **k** $\sqrt{7} - \sqrt{3}$ **l** $4 - 2\sqrt{2}$
m $9 - 15\sqrt{3}$ **n** $-14 - 14\sqrt{3}$ **o** $4 - 6\sqrt{2}$
- 3** **a** $4 + 3\sqrt{2}$ **b** $7 + 4\sqrt{3}$ **c** $1 + \sqrt{3}$ **d** $10 + \sqrt{2}$ **e** -2
f $3 - 3\sqrt{7}$ **g** $-1 - \sqrt{5}$ **h** 4 **i** 5 **j** $14 - 7\sqrt{2}$
- 4** **a** $3 + 2\sqrt{2}$ **b** $7 - 4\sqrt{3}$ **c** $7 + 4\sqrt{3}$ **d** $6 + 2\sqrt{5}$
e $5 - 2\sqrt{6}$ **f** $27 - 10\sqrt{2}$ **g** $9 + 2\sqrt{14}$ **h** $22 - 8\sqrt{6}$
i $8 - 4\sqrt{3}$ **j** $13 + 4\sqrt{10}$ **k** $13 - 4\sqrt{10}$ **l** $44 + 24\sqrt{2}$
m $51 - 10\sqrt{2}$ **n** $17 - 12\sqrt{2}$ **o** $19 + 6\sqrt{2}$
- 5** **a** 13 **b** 23 **c** 1 **d** -9 **e** 14
f 19 **g** -2 **h** -28 **i** -174
- 6** **a** 1 **b** -4 **c** $x - y$

Harder Questions

- 4** **a** -6 **b** $-14 - 2\sqrt{5}$ **c** $12 + 8\sqrt{2} - 9\sqrt{3} + 2\sqrt{6}$
- 5** $3 + 2\sqrt{2}$ **6** $\sqrt{6} = \frac{12}{5 - 6p}$ $\left(\text{or } \sqrt{6} = \frac{\sqrt{2} - \sqrt{3}}{p} \right)$
- 7** $x^2 = 8 - 2\sqrt{15}$ and so $2\sqrt{15} = 8 - x^2$
Now square both sides.
- 8** $u_1 = 1, u_2 = 1, u_3 = 2, u_4 = 3$